become accessible. Now, although it is hard to picture in more than three dimensions, this conclusion—more dimensions means more vibrational patterns—is general. If a string can vibrate in a fourth spatial dimension, it can execute more vibrational patterns than it could in only three; if a string can vibrate in a fifth spatial dimension, it can execute more vibrational patterns than it could in only four; and so on. This is an important realization, because there is an equation in string theory that demands that the number of independent vibrational patterns meet a very precise constraint. If the constraint is violated, the mathematics of string theory falls apart and its equations are rendered meaningless. In a universe with three space dimensions, the number of vibrational patterns is too small and the constraint is not met; with four space dimensions, the number of vibrational patterns is still too small; with five, six, seven, or eight dimensions it is still too small; but with nine space dimensions, the constraint on the number of vibrational patterns is satisfied perfectly. And that's how string theory determines the number of space dimensions.[36][19]

While this illustrates well the interplay of geometry and physics, their association within string theory goes further and, in fact, provides a way to address a critical problem encountered earlier. Recall that, in trying to make detailed contact between string vibrational patterns and the known particle species, physicists ran into trouble. They found that there were far too many massless string vibrational patterns and, moreover, the detailed properties of the vibrational patterns did not match those of the known matter and force particles. But what I didn't mention earlier, because we hadn't yet discussed the idea of extra dimensions, is that although those calculations took account of the *number* of extra dimensions (explaining, in part, why so many string vibrational patterns were found), they did not take account of the small size and complex *shape* of the extra dimensions—they assumed that all space dimensions were flat and fully unfurled—and that makes a substantial difference.

Strings are so small that even when the extra six dimensions are crumpled up into a Calabi-Yau shape, the strings still vibrate into those directions. For two reasons, that's extremely important. First, it ensures that the strings always vibrate in all nine space dimensions, and hence the constraint on the number of vibrational patterns continues to be